

MD. SAIDUZZAMAN

AI/ML Researcher | Computer
Science & Engineering Graduate



Personal details

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Skills

- Programming: Python, C++, Java, JavaScript
- Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Computer Vision, Explainable AI, Multimodal Learning
- AI/ML Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, SHAP
- Bioinformatics: AlphaFold, Protein Structure Analysis, Binding-Site Prediction, Computational Drug Discovery
- Web Development: React, FastAPI, JavaScript, HTML, CSS
- Databases: PostgreSQL, MySQL
- Tools & Deployment: Git, GitHub, Docker, Linux

Profile

Computer Science & Engineering graduate with a CGPA of 3.65/4.00 from Varendra University, with research and project experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Explainable AI. Experienced in applying AI and computational methods to healthcare, agriculture, and bioinformatics problems, with conference research contributions and a bachelor's thesis involving AlphaFold and machine learning. Interested in advanced AI research, particularly deep learning, computer vision, explainable AI, and multimodal learning.

Education

- Bachelor of Science (B.Sc.) in Computer Science & Engineering** Sep 2022 - Jul 2026
Varendra University, Rajshahi
 - CGPA: 3.65/4.00

ACADEMIC ACTIVITIES & ACHIEVEMENTS

- UCICS 2026 - Certificate of Participation
- UCICS 2026 - Certificate of Appreciation for Volunteer Contribution
- UCICS 2026 - Research Author Recognition
- UCICS 2025 - Research Author Recognition
- National AI Build-a-Thon - Final Round Participant, Milliox Bangladesh
- VU Tech Carnival 2024 - Certificate of Achievement
- 3D Printing Workshop - Robotic Society of Varendra University
- AI for NextGen Leaders - Varendra University Science Club
- Tech Debate 3.0 - Department of Computer Science & Engineering, Varendra University

RESEARCH EXPERIENCE

- Bachelor's Thesis Research Binding Site Prediction of Tuberculosis-Related Proteins Using AlphaFold and Machine Learning Approaches: A Computational Study**
 - Conducted computational research on binding-site prediction for tuberculosis-related proteins.
 - Investigated AlphaFold-derived protein structures and machine-learning approaches for identifying potential protein binding sites.
 - Applied computational methods relevant to bioinformatics and structure-based drug discovery.
 - Status: Thesis research completed; manuscript in preparation for journal submission.
 - github.com/Shimul42/tb-bindind-site
- Facial Emotion Recognition Deep Learning Research — Ongoing** Present
 - Investigating facial emotion classification using deep learning and computer vision techniques.
 - Working with image preprocessing, data augmentation, transfer learning, and multi-class classification & exploring deep learning approaches

Languages

Bengali	
English	
Hindi	

PUBLICATIONS & CONFERENCE PAPERS

Clinical Utility of Machine Learning Algorithms in Predicting the Presence of Heart Disease: A Cross-Validated Analysis

2026

UCICS 2026 — 2nd Undergraduate Conference on Intelligent Computing and Systems, Varendra University Status: Accepted, published, and presented.

- Research contribution on machine-learning approaches for predicting the presence of heart disease.
- Investigated predictive modeling using cross-validated machine-learning approaches.

Advancing Plastic Pollution Detection in Underwater Environments Using CNN Architectures

2025

UCICS 2025 — 1st Undergraduate Conference on Intelligent Computing and Systems, Varendra University Status: Accepted, published, and presented.

- Research contribution applying convolutional neural-network architectures to underwater plastic-pollution detection.
- Investigated computer-vision approaches for automated detection of plastic pollution in underwater environments.

SELECTED PROJECT

AgroGuard AI 2.0 AI-Powered Smart Agriculture Advisory Platform

- Developed an AI-powered agricultural advisory platform integrating weather information and computer-vision-based crop disease detection.
- Implemented a modern web architecture using React for the frontend and FastAPI for the backend.
- Integrated machine-learning/deep-learning components for agricultural decision support.
- Used PostgreSQL for data management and Docker for deployment.
- Technologies: Python, Machine Learning, Deep Learning, Computer Vision, React, FastAPI, PostgreSQL, Docker
- GitHub: github.com/Shimul42/AgroGuard-AI-2.0

RESEARCH INTERESTS

■ Artificial Intelligence

■ Deep Learning

■ Explainable AI

■ AI for Healthcare

■ AI Applications in Bioinformatics

■ Protein Structure Analysis

■ Machine Learning

■ Computer Vision

■ Multimodal Machine Learning

■ AI for Agriculture

■ Computational Biology